Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Loops, classes, lists, if statements, print, functions

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Easiest the sketch of the design, Hardest the coding

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Coding the whole thing is gonna take most of them time and effort

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Games, Friends and I will minimize them by going to a quiet place. I will need AI help, and I have no questions about the assignment.

1) Site Diagram

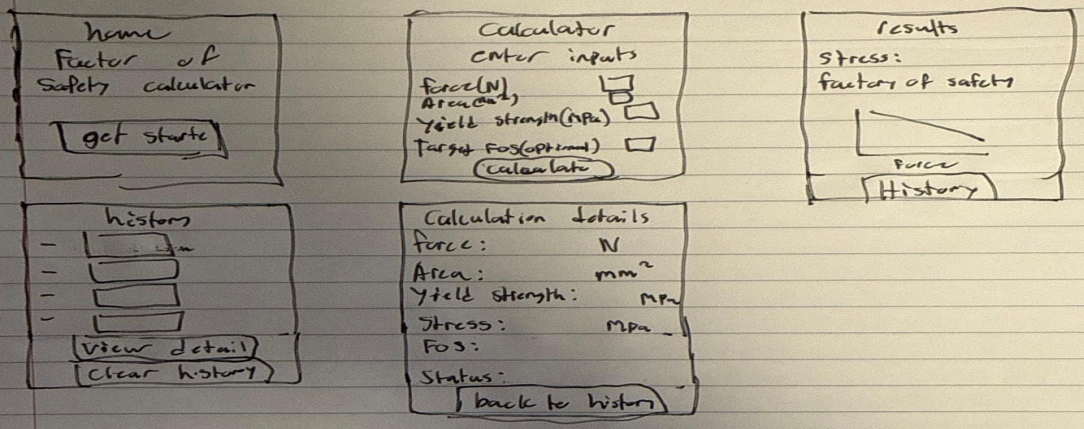
Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Mock design


Name
Factor of
Safety calculator
get starte
Calculator
enter inputs
force(N),
Area(mm²),
Yield strength(MPa)
Target FOS(optional)
calculate
results
Stress:
factory of safety
Force
History
history
view detail
Clear history
Calculation details
force:
N
Area:
mm²
Yield strength:
MPa
Stress:
MPa
FOS:
Status:
back to history

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**It is a calculator to help calculate normal stress and a factor of safety.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	1. Force_input(str)-stores applied force 2. Area_input(str)-stores cross-sectional area 3. Yield_strength(str)-store material strength 4. Target_fos_input(str)-stores users factor of safety 5. calculations(list of calculation objects)-stores previous calculations
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Calculation 1. Force(float)-applied force 2. Area(float)-cross-sectional area in square mm 3. Yield_strength(float)-material yield 4. Stress(float)-calculated normal stress 5. factor_of _safety(float)-calculate factor of safety 6. Target_fos(float)-users desired safety 7. assessment(str)-a message if met with user target safety
Routes (minimum 7)	1. Home-displays welcome message 2. Calculator-displays input 3. Calculate-processes the users input

The names of your route functions	4. Results-displays the calculated stress and safety 5. History-displays previous calculations 6. Calculation_details-displays the details of a selected previous calculation 7. Clear_history-clears saved calculation history
Pages (minimum 5) A brief description of the pages in your site	1. Home page-introduces application 2. Calculator page-allows users to enter force, cross-sectional area, yield strength, target safety 3. Results page-displays results of calculations 4. History page-shows list of previous calculations 5. Calculation details page-shows the inputs and results for calculation 6. About page-explains purpose of calculator
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	1. Textbox- Applied Force:Allows users to enter the applied force in newtons 2. Texbox- Cross-Sectional Area:Allows users to enter components of area 3. Textbox- Yield Strength:Allows users to enter material's yield strength 4. Textbox- Target Factor of Safety:Allows users to enter their desired minimum factor of safety
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. Check whether the user entered the valid positive number before calculating 2. Check whether the cross-sectional area in greater than zero to prevent division by zero 3. Compare the calculated factor of safety with the users target safety 4. Check whether calculation history is empty before displaying previous calculations.
Loop (minimum 1) This loop must iterate over the	I will use a for loop to iterate through the list of calculation objects stored in the State dataclass. The loop will display each previous calculations applied

list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	force, normal stress, and factor of safety on the history page.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.